

M0 · Dummy baseline

Model 1 of the ladder · the floor every real model must beat · AI Credit Risk / Loan Approval Engine

In one sentence: M0 ignores every feature and predicts the same probability of default for **every** applicant: the training default rate, **17.24%**. It cannot tell a safe borrower from a risky one, and that is exactly the point: it shows what "zero intelligence" scores, so we can measure how much each real model adds.

How it works

- **Training** = counting. It looks only at the training labels (388,124 loans, 66,928 defaults) and stores the share of defaults. No features, no fitting of weights or trees.
- **Prediction** = returning that stored number. `predict_proba` gives $[P(\text{repay}), P(\text{default})] = [0.8276, 0.1724]$ for everyone.
- **Versions A and B are identical:** B removes LendingClub's `grade` / `sub_grade` / `int_rate` / `installment`, but M0 uses no features at all.

Parameters

Parameter	Value used	What it means
strategy	"prior"	Predict the class frequencies seen in training as probabilities. Chosen because it gives honest probabilities (PD = 17.24%), which the probability metrics (Brier, log loss) need.
constant	None (unused)	Only used by strategy="constant" (always predict one fixed class).
random_state	None (unused)	Only used by the random strategies ("stratified", "uniform").

Other strategy	Why not used
most_frequent	Same predicted class (repay) but its probabilities are 0 / 1, which break Brier and log loss.
stratified / uniform	Random guesses: results change with the seed and add noise without information.
constant	Forcing one class is the same accuracy trap as most_frequent.

Data and rules

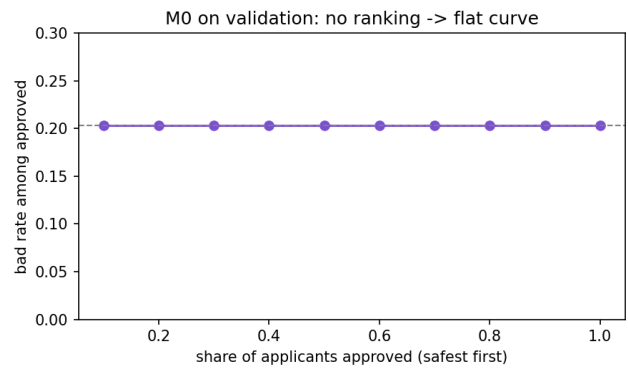
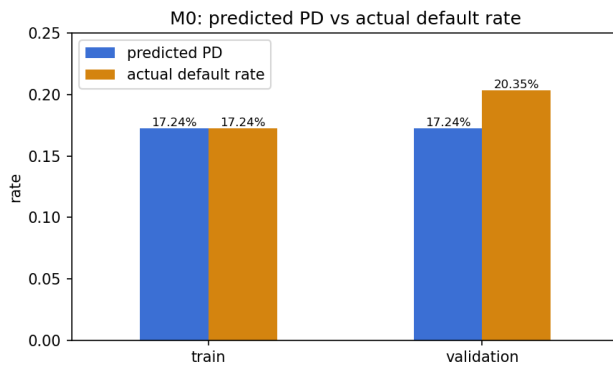
- Fitted on **train** (Aug-2012 to Dec-2014); reported on **train** and **validation** (2015 H1).
- The **test set stays sealed** until the final model is chosen; the 2016-2018 stress set is not used.
- Run in Jupyter: Python 3.13.9, pandas 2.3.3, scikit-learn 1.7.2.

Results

Split	Rows	Actual default rate	Predicted PD	ROC-AUC	KS	PR-AUC	Brier	Log loss
Train	388,124	17.24%	17.24%	0.500	0.000	0.1724	0.1427	0.4597
Validation	162,745	20.35%	17.24%	0.500	0.000	0.2035	0.1631	0.5085

What we learned

- **No ranking ability.** ROC-AUC 0.500 and KS 0.000. Approving the "safest" 10% or 90% of applicants gives the same 20.35% bad rate, a flat approval curve.
- **The accuracy trap.** M0 scores **79.6%** accuracy on validation while catching **0%** of defaults. This is why accuracy is never used to judge models in this project.
- **Drift.** The constant 17.24% under-predicts the real 20.35% in 2015 H1 by 3.11 points, so the final model must be calibrated on validation.
- **Loan term alone matters.** On validation, 60-month loans default at 35.40% vs 15.12% for 36-month loans, yet M0 gives both 17.24%. Even one feature would beat M0.



The bar M1 (Logistic Regression) must clear on validation: ROC-AUC above 0.500, KS above 0, PR-AUC above 0.204, Brier below 0.1631 and log loss below 0.5085. Beating M0 is expected; the size of the gap is what shows how much the features are worth.

Metric glossary. ROC-AUC: chance a random defaulter gets a higher PD than a random payer (0.5 = coin flip). KS: largest gap between the score distributions of defaulters and payers. PR-AUC: precision vs recall for defaults (baseline = default rate). Brier: mean squared error of the probabilities (lower is better). Log loss: penalty for confident wrong probabilities (lower is better).